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Week 1+2

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Exercise 1.1

A nonassociative algebra $(A, \circ)$ is called ***left symmetric*** if the associators are left symmetric in the sense that

$$(x, y, z) = (y, x, z) \quad ((x, y, z) := (x \circ y) \circ z - x \circ (y \circ z))$$

for all $x, y, z \in A$. Prove that for every left symmetric algebra $(A, \circ)$, $(A, [\cdot, \cdot])$ is a Lie algebra, where

$$[\cdot, \cdot] : A \times A \to A, \quad (x, y) \mapsto [x, y] := x \circ y - y \circ x$$

denotes the commutator on A.

Proof. To show $[\cdot, \cdot]$ is bilinear, we see

$$\begin{aligned}[x + y, z] &= (x + y) \circ z - z \circ (x + y) \\ &= (x \circ z - z \circ x) + (y \circ z - z \circ y) \\ &= [x, y] + [y, z]\end{aligned}$$

Similarly

$$\begin{aligned}[x, y + z] &= x \circ (y + z) - (y + z) \circ x \\ &= (x \circ y - y \circ x) + (x \circ z - z \circ x) \\ &= [x, y] + [y, z]\end{aligned}$$

Futhermore, it is obvious that

$$[x, x] = x \circ x - x \circ x = 0$$

Then we only need to show the Jacobi identity. Since $(A, \circ)$ is left symmetric, we have

$$\begin{aligned}
&[x\,[yz]] + [y\,[zx]] + [z\,[xy]]\\
&= x \circ (y \circ z - z \circ y) - (y \circ z - z \circ y) \circ x\\
&+ y \circ (z \circ x - x \circ z) - (z \circ x - x \circ z) \circ y\\
&+ z \circ (x \circ y - y \circ x) - (x \circ y - y \circ x) \circ z\\
&= x \circ (y \circ z) - x \circ (z \circ y) - (y \circ z) \circ x + (z \circ y) \circ x\\
&+ y \circ (z \circ x) - y \circ (x \circ z) - (z \circ x) \circ y + (x \circ z) \circ y\\
&+ z \circ (x \circ y) - z \circ (y \circ x) - (x \circ y) \circ z + (y \circ x) \circ z\\
&= (x \circ (y \circ z) - (x \circ y) \circ z) - (x \circ (z \circ y) + (x \circ z) \circ y)\\
&- ((y \circ z) \circ x + y \circ (z \circ x)) + ((z \circ y) \circ x - z \circ (y \circ x))\\
&- (y \circ (x \circ z) - (y \circ x) \circ z) - ((z \circ x) \circ y - z \circ (x \circ y))\\
&= 0
\end{aligned}$$

Thus $(A, [\cdot,\cdot])$ is a Lie algebra. □

Exercise 1.2

Write $\mathrm{Der}(A)$ for the Lie algebra of derivations on A, where $A = \mathbb{C}[t, t^{-1}]$. Prove that $\mathrm{Der}(A)$ has a basis $\{L_n := t^{n+1}\frac{d}{dt} \mid n \in \mathbb{Z}\}$ and the Lie brackets among the basis elements are as follows:

$$[L_n, L_m] = (m-n)L_{m+n} \quad (m, n \in \mathbb{Z}).$$

Proof. Let $D \in \mathrm{Der}\,(A)$ be any derivation on A. Then

$$\begin{aligned}
D\,(1) &= D\left(t \cdot t^{-1}\right)\\
&= D\,(t) \cdot t^{-1} + tD\left(t^{-1}\right)
\end{aligned}$$

The other hand

$$D\,(1) = D\,(1 \cdot 1) = D\,(1) \cdot 1 + 1 \cdot D\,(1) = 2D\,(1) \implies D\,(1) = 0$$

Thus we have

$$D\,(t) \cdot t^{-1} + tD\left(t^{-1}\right) = 0 \implies D\left(t^{-1}\right) = -t^{-2}D\,(t)$$

From the linearity and the product rule, we observe that a dirivation $D \in \mathrm{Der}\,(A)$ is totally determined by its value on t, that is for any $D_1, D_2 \in \mathrm{Der}\,(A)$ if $D_1\,(t) = D_2\,(t)$, then $D_1 = D_2$. It is straightforward to verify that L_n is a derivation on A for each $n \in \mathbb{Z}$. Note that

$$L_n(t) = t^{n+1}, \quad n \in \mathbb{Z}.$$

Since $D(t) \in A$, we have

$$D(t) \in \operatorname{span}(\cdots, L_{-2}(t), L_{-1}(t), L_0(t), L_1(t), L_2(t), \cdots).$$

Then we get

$$D \in \operatorname{span}(\cdots, L_{-2}, L_{-1}, L_0, L_1, L_2, \cdots).$$

Suppose we have a finite linear combination of the L_n that equals the zero derivation: $\sum_{n\in I} k_n L_n = 0$, where $I \subset \mathbb{Z}$ is a finite set and $k_n \in \mathbb{C}$. To check whether this is the zero derivation, we can test its action on the generator t. $\left(\sum_{n\in I} k_n L_n\right)(t) = 0(t) = 0$. This gives: $\sum_{n\in I} k_n L_n(t) = \sum_{n\in I} k_n t^{n+1} = 0$. The set of monomials $\{t^j\}_{j\in\mathbb{Z}}$ is a basis for A as a $\mathbb{C}$-vector space, so they are linearly independent. The equation $\sum_{n\in I} k_n t^{n+1} = 0$ implies that all coefficients must be zero, i.e., $k_n = 0$ for all $n \in I$. This proves that the set $\{L_n\}_{n\in\mathbb{Z}}$ is linearly independent. Combining above all , we know $\operatorname{Der}(A)$ has a basis $\{L_n : n \in \mathbb{Z}\}$.

To show $[L_n, L_m] = (m-n) L_{m+n}$, we only need to show that they agree on t:

$$\begin{aligned}
[L_n, L_m](t) &= (L_n(L_m(t)) - L_m(L_n(t))) \\
&= \left(L_n\left(t^{m+1}\right) - L_m\left(t^{n+1}\right)\right) \\
&= \left((m+1)\, t^{m+n+1} - (n+1)\, t^{m+n+1}\right) \\
&= (m-n)\, t^{m+n+1} \\
&= (m-n)\, L_{m+n}(t)
\end{aligned}$$

which completes the proof.

□

Exercise 1.3

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and let V be an ***n-dimensional*** vector space over $\mathbb{F}$ equipped with a nondegenerate bilinear form

$$B : V \times V \to \mathbb{F}, \quad (v, w) \mapsto B(v, w).$$

Prove that:

(a) If $\mathbb{F} = \mathbb{C}$ and B is symmetric, then there exists a basis $\{v_1, v_2, \dots, v_n\}$

of V such that $B(v_i, v_j) = \delta_{i,j}$ for all $1 \le i, j \le n$.

(b) If $\mathbb{F} = \mathbb{R}$ and B is symmetric, then there exists a basis $\{v_1, v_2, \dots, v_n\}$ of V and $p = 0, 1, \dots, n$ such that $B(v_i, v_j) = \varepsilon_i \delta_{i,j}$ for all $1 \le i, j \le n$, where $\varepsilon_i = 1$ if $i \le p$ and $\varepsilon_i = -1$ if $i > p$.

(c) If $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and B is skew-symmetric, then $n = 2m$ is even and there exists a basis $\{v_1, v_2, \dots, v_n\}$ of V such that the matrix

$$(B(v_i, v_j))_{1 \le i,j \le n} = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix},$$

where I_m denotes the identity matrix of rank m.

Proof. Lemma: If V is a vector space on the field $\mathbb{K}$ satisfying $\operatorname{char}(K) \neq 0$, B is a nonzero symmetric bilinear form on V , then there exists $v \in V$ such that $B(v, v) \neq 0$.

Proof of Lemma: Otherwise, we have for each $w, v \in V$,

$$\begin{aligned} 0 &= B(w + v, w + v) \\ &= B(w, w) + B(w, v) + B(v, w) + B(v, v) \\ &= 2B(w, v) \end{aligned}$$

which leads to a contradiction since B is nondegenerate.

(a) We induct on the dimension n. If $n = 1$, there exists vector v_0 such that $B(v_0, v_0) \neq 0$. We denote $B(v_0, v_0)$ by $\lambda \in \mathbb{C}$. λ has a square root in C, denoted by $\sqrt{\lambda}$. Let $v_1 = \frac{v_0}{\sqrt{\lambda}}$. It constitute a basis of V such that

$$B(v_1, v_1) = \frac{1}{\lambda} B(v_0, v_0) = 1$$

Now we suppose that $(n-1)$-dimensional space has property (a). We consider the n-dimensional vector space V on $\mathbb{C}$, with a nondegenerate bilinear form B. By the Lemma, we can take a vector v_0 such that $B(v_0, v_0) \neq 0$, and $v_1 := \frac{v_0}{\sqrt{B(v_0, v_0)}}$ such that $B(v_1, v_1) = 1$. Now we define

$$W := \{w : B(v_1, w) = 0\},$$

which is a subspace of V. We claim that $V = W \oplus \langle v_1 \rangle$, and thus $\dim W = n - 1$.

1. $W \cap \langle v_1 \rangle = 0$: Take any $w \in W \cap \langle v_1 \rangle$, then there is a complex valued c such that $w = cv_1$. Futhermore,

$$0 = B(w, v_1) = cB(v_1, v_1) = c \implies c = 0$$

There must be $w = 0$.

2. $W + \langle v_1 \rangle = V$: Take any $v \in V$. Let $w = v - B(v, v_1)v_1$, then

$$B(w, v_1) = B(v - B(v, v_1)v_1, v_1) = B(v, v_1) - B(v, v_1) = 0$$

Thus $w \in W$, $v = w + B(v, v_1)v_1$. Therefore $v \in W + \langle v_1 \rangle$.

Now we prove that $B|_W : W \times W \to \mathbb{C}$, $B|_W(w_1, w_2) := B(w_1, w_2)$ is a nondegenerate bilinear form on W.

1. Nondegenerate: For $w_0 \in W$, if for any $w \in W$, there is $B|_W(w_0, w) = 0$, then for any $v \in V$, set $v = k_1 v_1 + k_2 w_1$, $k_1, k_2 \in \mathbb{C}, w_1 \in W$. We have

$$B(w_0, v) = k_1 B(w_0, v_1) + k_2 B(w_0, w_1) = 0 + 0 = 0$$

Since B is nondegenerate, $w_0 = 0$. Therefore $B|_W$ is nondegenerate.

2. Symmetric and bilinear: Since B is symmetrc and bilinear, it is obvious that $B|_W$ is as well.

From our inductive assumption, ther exists a basis $\{v_2, \cdots, v_n\}$ for W such that $B\left(v_i, v_j\right) = \delta_{i,j}, 2 \le i, j \le n$. Then $\{v_1, \cdots, v_n\}$ is a basis for V such that $B\left(v_i, v_j\right) = \delta_{i,j}, 1 \le i, j \le n$.

(b) We induct on the dimension n. As in the (a), if $n = 1$, there exists vector v_0 such that $B(v_0, v_0) \neq 0$. We still denote $B(v_0, v_0)$ by λ , but here $\lambda \in \mathbb{R}$. We set

$$v_1 = \frac{v_0}{\sqrt{|\lambda|}}$$

Then

$$B(v_1, v_1) = B\left(\frac{v_0}{\sqrt{|\lambda|}}, \frac{v_0}{\sqrt{|\lambda|}}\right) = \frac{\lambda}{|\lambda|} = \begin{cases} 1, & \lambda > 0 \\ -1, & \lambda < 0 \end{cases}$$

Now we suppose that any $(n-1)$-dimensional space has property (b). We consider the n-dimensional vector space V on $\mathbb{R}$. A smililar manner as above gives a vector u_1 such that $B(u_1, u_1) = \pm 1 =: \varepsilon_1$. Still , define

$$W := \{w : B(u_1, w) = 0\}$$

After a same process as in (a), we have $V = W \oplus \langle u_1 \rangle$, $\dim W = n-1$, $B|_W$ is a nondegenerate bilinear form on W. From our inductive assumption, there exists a basis $\{u_2, \cdots, u_n\}$ for W such that $B\left(u_i, u_j\right) = \varepsilon_i \delta_{i,j}, \varepsilon_i \in \{-1, 1\}, 2 \leq i, j \leq n$. Then $\{u_1, \cdots, u_n\}$ is a basis for V such that $B\left(u_i, u_j\right) = \varepsilon_i \delta_{i,j}, \varepsilon_i \in \{-1, 1\}, 1 \leq i, j \leq n$. Finally , we can relable $\{u_1, \cdots, u_n\}$ to be $\{v_1, \cdots v_n\}$ such that there exists $p \in \{0, 1, \cdots, n\}$ such that $B\left(v_i, v_j\right) = \varepsilon_i \delta_{i,j}$ for all $1 \leq i, j \leq n$, where $\varepsilon_i = 1$ if $i \leq p$ and $\varepsilon_i = -1$ if $i > p$.

(c) For each $v \in V$, $B(v, v) = -B(v, v) \implies B(v, v) = 0$ since $\operatorname{char}(\mathbb{F}) \neq 2$. Let V be vector space with arbitray positive dimension on $\mathbb{F}$. Claim that exists vector $v_0, w_0 \in V$ such that $B(v_0, w_0) \neq 0$. Otherwise, for each $w, v \in V$,

$$B(w, v) = B(v, w) = 0$$

, which is a contradiction to B is nondegenerate. We set

$$v_1 = v_0, \quad w_1 = \frac{w_0}{B(v_0, w_0)}$$

Then $B(v_1, w_1) = 1$. Define

$$W \coloneqq \{w : B(w, \operatorname{span}(v_1, w_1)) = 0\}$$

Then

1. $W \cap \operatorname{span}(v_1, w_1) = 0$: Take any $u \in W \cap \langle v_1, w_1 \rangle$, then there exists $m, n \in \mathbb{F}$ such that $u = mv_1 + nw_1$. Then

   $$B(u, v_1) = B(mv_1 + nw_1, v_1) = nB(w_1, v_1) = 0 \implies n = 0$$

   $$B(u, w_1) = B(mv_1 + nw_1, w_1) = mB(v_1, w_1) = 0 \implies m = 0$$

   Thus $u = 0$.

2. $W + \operatorname{span}(v_1, w_1) = V$: Take any $v \in V$, let $w = v - B(v, w_1) v_1 - B(v_1, v) w_1$. Then

   $$B(w, v_1) = B(v - B(v, w_1) v_1 - B(v_1, v) w_1, v_1) = B(v, v_1) + B(v_1, v) = 0$$

   $$B(w, w_1) = B(v - B(v, w_1) v_1 - B(v_1, v) w_1, w_1) = B(v, w_1) - B(v, w_1) = 0$$

   Thus we have $V = W \oplus \operatorname{span}(v_1, w_1)$. $\dim W = \dim V - 2$.

   Now we claim that the dimension of V must be even. Otherwise, once we get $V = W \oplus \operatorname{span}(v_1, w_1)$, we can do the same thing for W and $B|_W$, and get $W = W_1 \oplus \operatorname{span}(v_2, w_2)$. But after finitely many decomposition, we get

$$V = \operatorname{span}(v_1, w_1) \oplus \cdots \oplus \operatorname{span}(v_m, w_m) \oplus W_{m-1}$$

such that $\dim W_{m-1} = 1$. Now there is no vector v_m, w_m such that $B|_{W_{m-1}}(v_m, w_m) = B(v_m, w_m) > 0$, which is a contradiction. Thus there exists $m \in \mathbb{Z}_{>0}$ such that $\dim V = 2m$. We have shown above that there exists $v_1, \cdots, v_m, w_1, \cdots, w_m \in V$ such that

$$V = \operatorname{span}(v_1, w_1) \oplus \cdots \oplus \operatorname{span}(v_m, w_m)$$

And $B(v_i, w_i) = 1, 1 \leq i \leq m$. $B\left(v_i, w_j\right) = 0, i \neq j$. We relabel

$$\{v_1, \cdots, v_m, w_1, \cdots, w_m\}$$

to be

$$\{v_1, \cdots, v_m, v_{m+1}, \cdots, v_n\}$$

Then we have

$$B(v_i, v_{i+m}) = 1, \quad B(v_{i+m}, v_i) = -1, \quad i = 1, 2, \cdots, m$$

and

$$B\left(v_i, v_{j+m}\right) = 0, \quad 1 \leq i \neq j \leq m$$

Thus we have the matrix

$$(B(v_i, v_j))_{1 \leq i,j \leq n} = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}.$$

$\square$

Exercise 1.4

Let L be a Lie algebra. Prove that:

(a) If $\varphi : L \to L'$ is a homomorphism of Lie algebras, then

$$L/\ker\varphi \cong \operatorname{im}\varphi.$$

(b) If I and J are ideals of L such that $I \subset J$, then J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J.

(c) If I is an ideal of L and J is a subalgebra of L, then there is a natural isomorphism between $(I+J)/I$ and $J/(I \cap J)$.

Proof. (a) Define $\bar{\varphi} : L/\ker\varphi \to \operatorname{Im}\varphi$

$$\bar{\varphi}(x + \ker\varphi) \coloneqq \varphi(x)$$

First we show that $\bar{\varphi}$ is well-defined. We suppose that $x + \ker\varphi = y + \ker\varphi$, then there exists $z \in \ker\varphi$ such that $y - x = z$. We have

$$\bar{\varphi}(y + \ker\varphi) = \varphi(y) = \varphi(x + z) = \varphi(x) + \varphi(z) = \varphi(x) = \bar{\varphi}(x + \ker\varphi)$$

Take any $x, y \in L$, $k \in F$. Then

$$\begin{aligned} \bar{\varphi}((x + \ker\varphi) + k(y + \ker\varphi)) &= \bar{\varphi}(x + ky + \ker\varphi) \\ &= \varphi(x + ky) \\ &= \varphi(x) + k\varphi(y) \\ &= \bar{\varphi}(x + \ker\varphi) + k\bar{\varphi}(y + \ker\varphi) \end{aligned}$$

Thus $\bar{\varphi}$ is a linear transformation. Furthermore,

$$\begin{aligned} \bar{\varphi}([x + \ker\varphi, y + \ker\varphi]) &= \bar{\varphi}([x, y] + \ker\varphi) \\ &= \varphi([x, y]) \\ &= [\varphi(x), \varphi(y)] \\ &= [\bar{\varphi}(x + \ker\varphi), \bar{\varphi}(y + \ker\varphi)] \end{aligned}$$

Thus $\bar{\varphi}$ is a homomorphism. We know from the definition of $\bar{\varphi}$ that $\bar{\varphi}$ is an epimorphism. To show that $\bar{\varphi}$ is a monomorphism, we suppose that $\bar{\varphi}(x + \ker\varphi) = \bar{\varphi}(y + \ker\varphi)$, then $\varphi(x) = \varphi(y)$. We have

$$\varphi(y - x) = \varphi(y) - \varphi(x) = 0 \implies (y - x) \in \ker\varphi \implies x + \ker\varphi = y + \ker\varphi$$

Thus $\bar{\varphi}$ is a monomorphism. Finally, we know $\bar{\varphi}$ is a isomorphism, then $L/\ker\varphi \simeq \operatorname{Im}\varphi$.

(b) First we show t hat J/I is a subspace of L/I. Take any element $x + I, y + I$ in J/I, and take any $k \in F$. We have

$$(x + I) + k(y + I) = (x + ky) + I \in J/I$$

since J is a subspace of L. Then, take any elment $x + I \in L/I$ and $y + I \in J/I$. Then

$$[x + I, y + I] = [x, y] + I \in J/I$$

since J is an ideal of L. We have J/I is an ideal of L/I. Define a map

$\psi : (L/I) / (J/I) \to L/J$

$$\psi((x+I)+(J/I)) \coloneqq x+J$$

If

$$(x+I)+J/I = (y+I)+J/I$$

Then

$$(x-y)+I \in J/I \implies x-y \in J \implies x+J = y+J$$

Thus ψ is well-defined. Then take any $k \in F$, we have

$$\begin{aligned}\psi(((x+I)+J/I)+k((y+I)+J/I)) &= \psi(((x+ky)+I)+J/I) \\ &= x+ky+J \\ &= (x+J)+k(y+J) \\ &= \psi((x+I)+J/I)+k\psi((y+I)+J/I)\end{aligned}$$

Thus ψ is a linear transformation. Furthermore,

$$\begin{aligned}\psi([(x+I)+J/I,(y+I)+J/I]) &= \psi(([x,y]+I)+J/I) \\ &= [x,y]+J \\ &= [x+J,y+J] \\ &= [\psi((x+I)+J/I),\psi((y+I)+J/I)]\end{aligned}$$

Thus ψ is a homomorphism. It is obvious that ψ is a epimorphism by definition. To show ψ is a monomorphism. We see that

$$x+J = 0 \iff x \in J \iff x+I \in J/I \iff (x+I)+(J/I) = (0+I)+(J/I)$$

Thus $\ker\psi = 0$, ψ is a monomorphism. Finally, we know ψ is a isomorphism between $(L/I) / (J/I)$ and L/J.

(c) Since $[I,J] \subseteq I$, we have

$$[I+J,I] \subseteq [I,I]+[J,I] \subseteq I$$

Therefore I is an ideal of $I+J$. Furthermore,

$$[J,I\cap J] \subseteq [I,J] \cap [J,J] \subseteq I \cap J$$

Thus $I\cap J$ is an ideal of J. We define $\psi : (I+J)/I \to J/(I\cap J)$ as follows:

For each $(i+j)+I \in (I+J)/I$, where $i \in I, j \in J$, we define

$$\psi((i+j)+I) := j + (I \cap J)$$

If $i_1+j_1+I = i_2+j_2+I$, by substracting j_1+i_2+I both sides, then $i_1-i_2 = j_2-j_1+I \implies j_2-j_1 \in I$. Since $j_2-j_1 \in J$, $j_1+(I\cap J) = j_2+(I\cap J)$, which shows that ψ is well-defined. To show ψ is a homomorphism, take any $(i_1+j_1)+I, (i_2+j_2)+I \in (I+J)+I, k \in F$. We have

$$\begin{aligned}
\psi(((i_1+j_1)+I)+k((i_2+j_2)+I)) &= \psi(((i_1+ki_2)+(j_1+kj_2))+I) \\
&= (j_1+kj_2)+I\cap J \\
&= (j_1+I\cap J)+k(j_2+I\cap J) \\
&= \psi((i_1+j_1)+I)+k\psi((i_2+j_2)+I)
\end{aligned}$$

Thus ψ is a linear transformation. Furthermore,

$$\begin{aligned}
\psi([(i_1+j_1)+I,(i_2+j_2)+I]) &= \psi([i_1+j_1,i_2+j_2]+I) \\
&= \psi(([i_1,i_2]+[i_1,j_2]+[i_2,j_1])+[j_1,j_2]+I) \\
&= [j_1,j_2]+I\cap J \\
&= [\psi((i_1+j_1)+I), \psi(i_2+j_2)+I]
\end{aligned}$$

Thus ψ is a homomorphism. It is obvious that ψ is surjective by definition. To show that it is a monomorphism, note that

$$\psi((i+j)+I) = 0 \iff j+(I\cap J) = 0 \iff j \in I\cap J \iff (i+j) \in I$$

Thus $\ker\psi = 0$, which shows that ψ is an monomorphism. Finally, we know ψ is a isomorphism between $(I+J)/I$ and $J/(I\cap J)$

$\square$

Exercise 1.5

Let L be a Lie algebra, I an ideal of L, and $\pi : L \to L/I$ the quotient map. Prove that for any Lie homomorphism $\varphi : L \to L'$ satisfying that $I \subset \ker\varphi$, there is a unique homomorphism $\psi : L/I \to L'$ making the diagram

$$\begin{tikzcd}
L \arrow[r, "\varphi"] \arrow[d, "\pi"] & L' \\
L/I \arrow[ur, "\psi"] &
\end{tikzcd}$$

commute.

Proof. Since the diagram commutes, for each $x+I \in L/I$, the only choic of $\psi(x+I)$ is

$$\psi(x+I) = \varphi(x)$$

Now we show that such $\psi : L/I \to L'$ is well-defined and indeed a homomorphism, and thus the unique one. For which , suppose that $x_1+I = x_2+I$, then $x_1 - x_2 \in I \subseteq \ker\varphi$, we have

$$\psi(x_1+I) = \varphi(x_1) = \varphi(x_1)+\varphi(x_2-x_1) = \varphi(x_1+(x_2-x_1)) = \varphi(x_2) = \psi(x_2+I)$$

Thus ψ is well-defined. Then take any $x+I, y+I \in L/I, k \in F$, we have

$$\begin{aligned}\psi((x+I)+k(y+I)) &= \psi((x+ky)+I)\\ &= \varphi(x+ky)\\ &= \varphi(x)+k\varphi(y)\\ &= \psi(x+I)+k\psi(y+I)\end{aligned}$$

Thus ψ is a linear transformation. Furthermore,

$$\begin{aligned}\psi([x+I,y+I]) &= \psi([x,y]+I)\\ &= \varphi([x,y])\\ &= [\varphi(x),\varphi(y)]\\ &= [\psi(x+I),\psi(y+I)]\end{aligned}$$

Thus ψ is a homomorphism. □

Week 3

Exercise 2.1

Let $n \geq 2$ be a positive integer, and write $M_n(\mathbb{C})$ for the associative algebra consisting of all $n \times n$ complex matrices. Prove that $M_n(\mathbb{C})$ is simple.

Exercise 2.2

Let $n \geq 2$ be a positive integer, and write $\mathfrak{sl}_n(\mathbb{C})$ for the Lie algebra consisting of all trace zero $n \times n$ complex matrices. Prove that $\mathfrak{sl}_n(\mathbb{C})$ is simple.

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